

Shlok Noval

+91-9588621416 | [Email](#) | [Linkedin](#) | [GitHub](#) | [Portfolio](#) | Chh. Sambhajinagar (M.S.), India

Introduction

A highly motivated Computer Science Engineering student at JNEC, MGM University, currently in 3rd Year with Minors in Machine Learning & Artificial Intelligence. Passionate about solving real-world problems, AIML development, and gaining hands-on experience in this dynamic field.

Technical Skills

- **Programming Languages:** Intermediate: C++, C, Python, Dart, Java, Javascript
- **Frameworks & Platforms:** Flutter, Streamlit, Qt Creator, Arduino IDE, Node.js, Express.js, FastAPI, Socket.io, React.js
- **AI, GenAI, Machine Learning & Computer Vision:** LLaMA 3, Mistral, Ollama - Offline LLMs, YOLOV8, API's integration, Sentiment analysis, NLP, OpenCV
- **Cloud & API Integration:** Google Cloud Platform, Google Maps API, Firebase, Building & API testing with Postman
- **Databases:** MySQL, PostgreSQL, SQLite, Firestore, Supabase
- **Web & UI Technologies:** HTML5, CSS3, Web Design (MKCL Certified), UI/UX, Next.js
- **Tools & Environments:** Microsoft Office, GitHub, Visual Studio Code, Android Studio, Postman, XAMPP, MATLAB, Ubuntu Linux, Vercel, Netlify
- **Networking & Security:** Cyber Security, Networking, and IT Support (MKCL Certified)
- **Soft Skills:**
 - Hackathon Leadership & Project Ownership
 - Team Collaboration & Agile Workflow
 - Time Management & Deadline Efficiency
 - Fluent Communication & Public Speaking

Experience

IIT Hyderabad

Machine Learning research Intern (Onsite)

Hyderabad, India

June 2025 – August 2025

- 2D Particle tracker using MATLAB for plotting trajectories.
- SPINODE – Neural SDE using Neural Network – Python, CUDA, Euler DE, Meilstein DE.
- Neural Network integration in Stochastic physics informed – neural ordinary differential equation at BME Department, IITH.

Freelance Full Stack Developer

Self-Employed

2025 – Present

- Developed and deployed production-ready web platforms for clients using React.js, Next.js, Supabase, Firebase, and Vercel.
- Successfully delivered and maintained platforms serving 300+ users/members in production environments.
- Handled complete project ownership including UI/UX, backend integration, deployment, testing, and client coordination.

Project Work

1. Annada - Farmers Assistant (Flagship)

Flutter, Google (Gemini) APIs, Postman API

- Major Multi-lingual project consisting of an android application to assist farmers.
- Consists of Google Firebase Auth security integrated, Vertex AI powered Chatbot for instant assistance.
- Fertilizer Calculator & Yield predictor with In-built Area input (Acres/Hectres) and Fertilizer Assistance with approx. result values (NPK Dosage).
- Yojna section providing quick access to government scheme, and real-time Weather & News system.

2. Tathya - An AI Powered Mis-Information & Fake Info Detector

Python, Streamlit, Mistral Latest, Ollama, Lang-Chain

- Fake & Mis-information detector using Machine learning models and artificial intelligence.
- Specifically designed for PIBs and Government Agencies.

- Included Sentiment Analysis and NLP.

3. 2D Particle Tracker

MATLAB, Image processing, OpenCV

- Developed a MATLAB-based GUI to track 2D particle motion frame-by-frame and extract precise nanometer-level displacement.
- Integrated automated trajectory plotting, centroid detection, and real-time visualization for research analysis.

4. GreenLink - Emergency vehicle reporting & traffic clearance system

HTML5, CSS3, Vanilla JS, Firebase Auth, Firestore

- Real-time coordination platform connecting emergency vehicles, traffic incharge, and on-duty police officers.
- Features role-based dashboards with live status tracking, duty toggles, and clearance management.
- Utilizes Firestore for real-time onSnapshot data syncing and Leaflet.js for live GPS tracking and mapping.

5. Innovate Hub - Connect & Collaborate

HTML, Tailwind CSS, Firebase (Auth + Firestore), JavaScript

- Built a project marketplace with authentication and project info system.
- Users can find projects, join team, send resumes, get accepted, track project status with real-time admin dashboard for detailed info.

6. ArthaSathi - Accessible Financial Authorization Platform

Next.js, Express, FastAPI, Firebase, Gemini 2.5 Flash

- Built an intelligence-driven financial consent platform leveraging Generative AI to simplify dense legal jargon and extract critical risks from complex documents.
- Users can upload documents (PDF/images) for instant OCR extraction, verify their understanding through auto-generated quizzes, and ask questions via a multi-lingual, voice-enabled AI chatbot.
- Implemented a secure consent pipeline featuring biometric voice signing, standard digital signatures, cryptographic ledger hashing, and interactive analytics dashboards.

7. DivyaDrishti - Smart City Road Hazard Detection Platform

Next.js, Node.js, ML, Express, Socket.io, FastAPI, YOLOv8n

- Built a real-time AI computer vision platform to detect potholes, vehicle accidents, and stray animals from live CCTV feeds.
- Features sub-50ms hazard detection, a geospatial heatmap dashboard for operators to track incidents, and automated instant field alerts via WhatsApp/SMS for emergency dispatch.

Education

B. Tech., Computer Science Engineering, Jawaharlal Nehru Engineering College, MGM University

Current CGPA: 9.10

Chh. Sambhajinagar, MH, India | 2023 – Early 2027

Deogiri College

11th: 92.5%, 12th: 60% (Computer Science: 90%)

Chh. Sambhajinagar, MH, India | 2021 – 2023

Podar International School (C.B.S.E.)

10th: 89.80%

Chh. Sambhajinagar, MH, India | 2016 – 2021

Coursework

- Artificial Intelligence
- Machine Learning
- Discrete Mathematics
- Database Management System
- DevOps
- C/Java/Python/Js/C++

Achievements

- **Microsoft Office Specialist (MOS)**, Microsoft India – Secured AIR 2 in year 2021 and qualified for consecutively 3 years; awarded Gold and Silver Medals at National level.
- **National Level I.D.E.A. Hackathon powered by E3L x HBSU** – 1st Rank – Won 50,000 INR & Goodies.
- **TechSprint 2026 Hackathon Special Sponsor Award powered by Google Developers Group OC** – 1st Rank – Won 50,000 INR worth Coupons & Internship.
- **National Level Hack The Gap 2.0 2026** – 2nd Runner Up – Won Trophy & 15,000 INR.
- **State Level Stellar – Supernova Hackathon** – 1st Rank – Won 5,000 INR & Goodies.

- **National Level Techathon 3.0 Hackathon Domain Winner** – 1st Rank – Won Trophy & 10,000 INR.
- **National Level MAGIC BATIC 2025-26** – Top 10 – PAN India.
- **1st Rank** – E-GOV Hackathon Pitch at Swayambhu 2025, MGM University.
- **3rd Rank** – SharpTech, MGM University.
- **6th Rank** – E-MERGE 2.0 Entrepreneurship Challenge 2024, IIT Hyderabad.
- **SIH 2025** – College Level Finalists.
- **Champion Milestone Achiever** – Google Arcade & Google Cloud Arcade Milestone Program 2024 and 2025.
- **State-level NCO Olympiad** Qualifier.
- **Completed 15+ certified courses from MKCL and Cisco** (Maharashtra Knowledge Corp. Ltd.), including: MS-CIT, Web Designing, C/C++ Programming, Desktop Publishing, Graphic Designing, Cisco: Cyber Security, and IT Hardware, Desktop & Network Support.
- **IIT Bombay Spoken tutorial:** Scored 90% in Intro to computers (Credit: 1), and 85% in Python 3.4.3 (Credits: 4).
- **Winner of Dance Competition** - Solo & Group dance 2025 and 2026 at Razzmatazz (Annual Fest) 2025, MGM University, Chh. Sambhajinagar.
- **TopMate Certified Mentor** – Mentored **100+** Students by conducting seminars and **1:1** guidance session.

Co-curricular & Leadership Involvement

- **Google Student Ambassador 2025** – Lead 2026-2027, JNEC, MGM University.
- **Event management Head** – Training and Placement 2025-2027, JNEC, MGM University.
- **Content Creation Team** – BlackBox AI Mavericks Club 2024-2025, JNEC, MGM University.
- **Event Management Team** – GDG (Google Developer Group) 2024-2025, MGM University.
- **Overall Event Head** – E3L Hackathon 2025, HBSU, Mumbai – Coordinated with 100+ Teams across India.
- **Official Logistics Team/ Logistics Head** – TEDx MGMU 2024 and 2025 – Managed logistics for event comprising 200+ Members.
- **Dance Choreographer** – Selected performer and cultural team lead during university induction – Choreographed 50+ Students.
- **NSS (National Service Scheme) Green Club Campaign Coordinator** – Led 50+ Student and Engaged in community outreach, camps, youth programs, Led awareness & sustainability initiatives from year 2024 - year 2025.
- **Internal Organizing Committee** – Smart India Hackathon 2025 – Managed 50+ Teams for SIH 2025.